

Paper 1 - Set 05 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Differentiation
2	D	Quadratic functions
3	E	Ratio, proportion and percentages
4	F	Number, units and estimation
5	A	Graphs of functions
6	F	Straight lines
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10	A	Polynomials, factor and remainder theorems
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Worked solutions

Question 1 DIF-10 · Differentiation · spec MM6.3

The curve $y = x^3 + px^2 + qx$ has stationary points at $x = 1$ and $x = 3$. Find q .

Answer: C

$$y' = 3x^2 + 2px + q = 3(x - 1)(x - 3) = 3x^2 - 12x + 9. \text{ So } q = 9 \text{ (and } p = -6\text{).}$$

Full method - how and why

Known stationary points mean known roots of f' . Compare the factorised derivative with its expanded form.

- 1 Differentiate: $y' = 3x^2 + 2px + q$.

why: A cubic's derivative is a quadratic.

- 2 The stationary points are its roots, so $y' = 3(x - 1)(x - 3)$.

why: The leading coefficient must be 3 to match $3x^2$.

- 3 Expand: $3(x^2 - 4x + 3) = 3x^2 - 12x + 9$.

why: Multiply out to compare coefficients.

- 4 Compare constants: $q = 9$ (and comparing x terms gives $2p = -12$, so $p = -6$).

why: The question asks only for q .

Where it goes wrong

- Forgetting the leading 3 and getting $q = 3$.
- Answering -6 , which is p .

Takeaway Stationary points are roots of f' : build f' in factorised form with the correct leading coefficient.

Question 2 QUA-19 · Quadratic functions · spec MM1.3

The expression $x^2 + 6x + c$ can be written as $(x + p)^2 + q$. Given that $q = -c$, find the value of c .

Answer: D

$$(x + 3)^2 + (c - 9), \text{ so } q = c - 9 = -c, \text{ giving } 2c = 9 \text{ and } c = \frac{9}{2}.$$

Full method - how and why

Completing the square symbolically, then imposing a relationship between the two constants. It is an equation in c , not in x .

- 1 Complete the square: $x^2 + 6x + c = (x + 3)^2 - 9 + c$.

why: Half of 6 is 3; subtracting 9 compensates for the +9 the square introduces.

- 2 Identify $p = 3$ and $q = c - 9$.

why: Compare with the required form $(x + p)^2 + q$ term by term.

- 3 Impose $q = -c$: $c - 9 = -c$.

why: The condition given in the question is a link between q and c , so substitute your expression for q .

- 4 Solve: $2c = 9$, so $c = \frac{9}{2}$.

why: A simple linear equation - just collect the c terms.

Where it goes wrong

- Writing $q = c + 9$ (sign error) and getting $c = -\frac{9}{2}$.
- Solving $c - 9 = c$ and concluding there is no solution.

Takeaway Complete the square with the constant left as a letter; the extra condition then becomes an ordinary equation in that letter.

Question 3 RAT-07 · Ratio, proportion and percentages · spec M3.5

A mixture contains sand and cement in the ratio 5 : 2. How much cement must be added to 35 kg of the mixture to make the ratio 5 : 3?

Answer: E

35 kg contains 25 kg sand and 10 kg cement. For 5 : 3 with 25 kg sand, cement must be 15 kg: add 5 kg.

Full method - how and why

A changing ratio where one quantity stays fixed. Anchor on the unchanged amount.

- 1 Split the 35 kg in the ratio 5 : 2: seven parts of 5 kg, so 25 kg sand and 10 kg cement.

why: $35 \div 7 = 5$ kg per part.

- 2 Only cement is added, so the sand stays at 25 kg.

why: Identify what does **not** change - that is the anchor for the new ratio.

- 3 In the new ratio 5 : 3, the 5 parts correspond to 25 kg, so one part is 5 kg and the cement must be $3 \times 5 = 15$ kg.

why: Scale the new ratio to the fixed quantity.

- 4 Cement to add: $15 - 10 = 5$ kg.

why: The question asks for the amount added, not the final amount.

Where it goes wrong

- Answering 15 kg (the new total cement) - an offered option.
- Rescaling the sand as well, as though both quantities changed.

Takeaway In 'add to change the ratio' problems, anchor on the quantity that stays the same.

Question 4 NUM-05 · Number, units and estimation · spec M2.8

Evaluate $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$, giving your answer in standard form.

Answer: F

$$\frac{24 \times 10^2}{6 \times 10^{-3}} = 4 \times 10^5.$$

Full method - how and why

Standard-form arithmetic: handle the numbers and the powers of ten separately, then renormalise.

- 1 Multiply the numerator: $(3 \times 8) \times 10^{4+(-2)} = 24 \times 10^2$.

why: Multiply the leading numbers, add the indices.

- 2 Divide by 6×10^{-3} : $\frac{24}{6} \times 10^{2-(-3)} = 4 \times 10^5$.

why: Divide the numbers, subtract the indices - and subtracting -3 adds 3.

- 3 Check the form: $1 \leq 4 < 10$ ✓, so no renormalising is needed.

why: Standard form requires the leading number to be between 1 and 10.

Where it goes wrong

- Getting 10^{-1} by adding instead of subtracting the -3 .
- Leaving 24×10^2 , which is not in standard form.

Takeaway Numbers multiply/divide; indices add/subtract. Then check $1 \leq a < 10$.

Question 5 GRA-09 · Graphs of functions · spec MM8.1

The graph of $y = \log_2 x$ is reflected in the line $y = x$. The resulting graph has equation

Answer: A

Reflection in $y = x$ gives the inverse function: $x = \log_2 y$, i.e. $y = 2^x$.

Full method - how and why

Reflection in $y = x$ produces the inverse function - swap x and y and re-solve.

- 1 Reflecting in $y = x$ swaps the roles of x and y : the new graph satisfies $x = \log_2 y$.

why: Every point (a, b) maps to (b, a) .

- 2 Rewrite in exponential form: $y = 2^x$.

why: $\log_2 y = x$ means $y = 2^x$ - logs and exponentials are inverse functions.

Where it goes wrong

- Choosing $y = \log_2(-x)$ (a y -axis reflection) or $-\log_2 x$ (an x -axis reflection).
- Thinking the inverse of a log is x^2 because of the base 2.

Takeaway Reflection in $y = x$ gives the inverse; $y = a^x$ and $y = \log_a x$ are mirror images in that line.

Question 6 LIN-02 · Straight lines · spec MM3.1

The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

Answer: F

Gradient $AB = \frac{2}{4} = \frac{1}{2}$. Gradient $AC = \frac{6}{k-1} = \frac{1}{2} \Rightarrow k - 1 = 12 \Rightarrow k = 13$.

Full method - how and why

Collinear points have equal gradients between any pair. Set two gradients equal and solve.

- 1 Gradient of AB : $\frac{4-2}{5-1} = \frac{2}{4} = \frac{1}{2}$.

why: Any two of the three points determine the line, so compute the gradient from the pair with known coordinates.

- 2 Gradient of AC : $\frac{8-2}{k-1} = \frac{6}{k-1}$.

why: Use the same starting point A for both gradients to avoid sign confusion.

- 3 Set them equal: $\frac{6}{k-1} = \frac{1}{2}$.

why: Collinear means the gradient between any pair is the same.

- 4 Cross-multiply: $k - 1 = 12$, so $k = 13$.

why: Check: from $B(5, 4)$ to $C(13, 8)$ the gradient is $\frac{4}{8} = \frac{1}{2} \checkmark$.

Where it goes wrong

- Forgetting the -1 and answering 12.
- Inverting the gradient and solving $\frac{k-1}{6} = \frac{1}{2}$, which gives $k = 4$.

Takeaway Collinearity is an equal-gradient condition; always verify with the third pair of points.

Question 7 POL-05 · Polynomials, factor and remainder theorems · spec MM1.6

The polynomial $p(x) = x^3 + ax^2 + bx + 4$ has $(x - 1)$ and $(x + 2)$ as factors. Find the third linear factor.

Answer: B

Write $p(x) = (x - 1)(x + 2)(x - r)$. The constant term is $(-1)(2)(-r) = 2r = 4$, so $r = 2$; the third factor is $(x - 2)$.

Full method - how and why

A cubic with two known factors has a third. The constant term alone pins it down - no division needed.

- 1 Write $p(x) = (x - 1)(x + 2)(x - r)$, where $(x - r)$ is the unknown factor.

why: The leading coefficient is 1, so the product of the three monic linear factors is the whole cubic.

- 2 Compare constant terms: the constant of the product is $(-1)(2)(-r) = 2r$, and $p(x)$ has constant term 4.

why: The constant term of a product of brackets is the product of their constant terms - much faster than expanding everything.

- 3 Solve: $2r = 4$, so $r = 2$ and the third factor is $(x - 2)$.

why: Check: $(-1)(2)(-2) = 4$ ✓.

Where it goes wrong

- Reading the third factor as $(x + 2)$ by forgetting the sign flip between the root r and the factor $(x - r)$.
- Trying to find a and b first - unnecessary work.

Takeaway Compare constant terms to find a missing factor: the product of the constants in the brackets equals the constant term.

Question 8 PRO-01 · Probability · spec M7.7

A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

Answer: D

$$P(RB) + P(BR) = \frac{4}{10} \cdot \frac{6}{9} + \frac{6}{10} \cdot \frac{4}{9} = \frac{48}{90} = \frac{8}{15}.$$

Full method - how and why

'Different colours' happens in two orders. Either add the two ordered cases or use the complement.

- 1 Red then blue: $\frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$.

why: After removing a red, 9 counters remain and all 6 blues are still there - **without replacement**.

- 2 Blue then red: $\frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$.

why: The same product, but the orders are distinct outcomes so both count.

- 3 Add: $\frac{48}{90} = \frac{8}{15}$.

why: Mutually exclusive cases add; then cancel by 6.

- 4 Check by complement: $P(\text{same}) = \frac{4}{10} \cdot \frac{3}{9} + \frac{6}{10} \cdot \frac{5}{9} = \frac{12+30}{90} = \frac{42}{90}$, and $1 - \frac{42}{90} = \frac{48}{90}$ ✓.

why: Two routes agreeing is strong confirmation.

Where it goes wrong

- Counting only one order, giving $\frac{4}{15}$ - an offered option.
- Keeping the denominator at 10 for the second draw ($\frac{12}{25}$), which is the **with replacement** answer.

Takeaway Without replacement: reduce both the relevant colour count and the total; and count both orders unless the question fixes one.

Question 9 SIM-06 · Simultaneous equations and inequalities · spec MM1.4

Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

Answer: F

$x = y + 1$; $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$. Solutions $(3, 2)$ with $x + y = 5$ and $(-2, -3)$ with $x + y = -5$. The sum is 0.

Full method - how and why

A product and a difference. Substituting the linear relation gives a quadratic with two solution pairs, and the question wants a total over both.

- From $x - y = 1$, write $x = y + 1$.
why: Substituting the linear equation keeps things to one variable.
- Substitute into $xy = 6$: $(y + 1)y = 6 \Rightarrow y^2 + y - 6 = 0$.
why: Expand before rearranging into standard form.
- Factorise: $(y + 3)(y - 2) = 0$, so $y = -3$ or $y = 2$.
why: Both are valid - nothing in the question restricts the sign.
- Find the pairs: $y = 2 \Rightarrow x = 3$, giving $x + y = 5$; $y = -3 \Rightarrow x = -2$, giving $x + y = -5$. Sum of these: 0.
why: The question asks for the sum of **all possible values of** $x + y$, so compute $x + y$ for each pair and add.

Where it goes wrong

- Finding only the positive pair and answering 5.
- Adding all four numbers $3 + 2 + (-2) + (-3)$ - which happens to be 0 too, but for the wrong reason; on another question that luck runs out.

Takeaway When a system has several solution pairs, compute the requested quantity for each pair and then combine.

Question 10 POL-11 · Polynomials, factor and remainder theorems · spec MM1.6

Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.

Answer: A

$(x + 1)(x - 2) = x^2 - x - 2$; multiply by $(2x + 3)$: the x^2 term is $3x^2 - 2x^2 = x^2$, coefficient 1.

Full method - how and why

Only one coefficient is wanted, so do not expand everything: collect just the ways of making x^2 .

- Multiply the first two brackets: $(x + 1)(x - 2) = x^2 - x - 2$.
why: Two brackets at a time keeps the bookkeeping small.
- To get x^2 from $(x^2 - x - 2)(2x + 3)$, pair x^2 with 3, and $-x$ with $2x$.
why: An x^2 term arises from (degree 2)×(degree 0) and (degree 1)×(degree 1) - list the pairings rather than expanding.
- Add the contributions: $3x^2 + (-x)(2x) = 3x^2 - 2x^2 = x^2$, so the coefficient is 1.
why: Only these two products can contribute.

Where it goes wrong

- Missing the $(-x)(2x)$ contribution and answering 3.
- Expanding everything under time pressure and losing a sign.

Takeaway For a single coefficient, list the pairs of terms whose degrees add to the one you want.

Question 11 TRI-09 · Triangles, radians and sectors · spec MM4.1

Two sides of a triangle are 3 and 5 and the included angle is θ , where $\cos \theta = -\frac{1}{5}$. Find the length of the third side.

Answer: C

$$c^2 = 9 + 25 - 2 \cdot 3 \cdot 5 \cdot \left(-\frac{1}{5}\right) = 34 + 6 = 40.$$

Full method - how and why

Cosine rule with a negative cosine. The only difficulty is handling the double negative.

- 1 Write the rule: $c^2 = 3^2 + 5^2 - 2(3)(5)\cos \theta$.
why: Two sides and the included angle again.
- 2 Substitute $\cos \theta = -\frac{1}{5}$: $c^2 = 9 + 25 - 30\left(-\frac{1}{5}\right) = 34 + 6$.
why: Subtracting a negative adds: $-30 \times \left(-\frac{1}{5}\right) = +6$.
- 3 So $c^2 = 40$ and $c = \sqrt{40}$.
why: Leave it in surd form; $\sqrt{40} = 2\sqrt{10}$ if you prefer.

Where it goes wrong

- Getting $34 - 6 = 28$ by mishandling the sign, which is offered as $\sqrt{28}$.
- Forgetting the angle is obtuse, so the third side must be **longer** than $\sqrt{34}$ - a useful sanity check.

Takeaway A negative cosine makes the third side longer than Pythagoras would give; check your answer against $\sqrt{a^2 + b^2}$.

Question 12 QUA-02 · Quadratic functions · spec MM1.3

By completing the square, find the minimum value of $2x^2 - 12x + 23$.

Answer: B

$$2x^2 - 12x + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5, \text{ so the minimum is } 5 \text{ (at } x = 3\text{).}$$

Full method - how and why

Minimum value of a quadratic: complete the square. The leading coefficient 2 must be factored out **before** completing the square.

- 1 Factor the 2 out of the x terms: $2(x^2 - 6x) + 23$.
why: Completing the square only works on $x^2 + (\text{something})x$; the coefficient of x^2 must be 1 inside the bracket.
- 2 Complete the square inside: $x^2 - 6x = (x - 3)^2 - 9$, so the expression is $2[(x - 3)^2 - 9] + 23$.
why: Halve the coefficient of x ($-6 \rightarrow -3$) and subtract the square of that half.
- 3 Multiply out the 2: $2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$.
why: The -9 is inside the bracket, so it is multiplied by 2 as well - this is where marks are lost.
- 4 $2(x - 3)^2 \geq 0$, so the least value is 5, at $x = 3$.
why: A square is never negative, and it is zero exactly at the vertex.

Where it goes wrong

- Forgetting to multiply the -9 by 2, giving $2(x - 3)^2 + 14$ and the answer 14.
- Quoting the x -coordinate 3 instead of the minimum value 5 - both are offered.

Takeaway Take the leading coefficient outside first, complete the square, then multiply back in; the constant left outside is the minimum.

Question 13 GRA-01 · Graphs of functions · spec MM8.6

How many times does the graph of $y = x^3 - x^2 - 6x$ cross or touch the x -axis?

Answer: A

$$x(x^2 - x - 6) = x(x - 3)(x + 2); \text{ three distinct real roots, so three crossings.}$$

Full method - how and why

Axis crossings are roots. Factorise fully and count **distinct** roots.

1 Factor out x : $x^3 - x^2 - 6x = x(x^2 - x - 6)$.

why: There is no constant term, so x is a factor - always check this first.

2 Factorise the quadratic: $x^2 - x - 6 = (x - 3)(x + 2)$.

why: Two numbers multiplying to -6 , adding to -1 .

3 Roots: $x = 0, 3, -2$ - three distinct values, each a simple root, so the graph crosses three times.

why: Distinct simple roots mean genuine crossings; a repeated root would be a touch.

Where it goes wrong

- Missing the root $x = 0$ and answering 2.
- Counting the y -intercept as an extra crossing of the x -axis.

Takeaway Factor out x when there is no constant term; count distinct roots for the number of axis meetings.

Question 14 INT-12 · Integration · spec MM7.2

Find the value of $\int_1^8 x^{-\frac{2}{3}} dx$.

Answer: C

$$[3x^{1/3}]_1^8 = 3(2 - 1) = 3.$$

Full method - how and why

A negative fractional index. Add one to it, and choose limits that make the cube root easy.

1 Add one to the index: $-\frac{2}{3} + 1 = \frac{1}{3}$, and divide by $\frac{1}{3}$ (i.e. multiply by 3): $\int x^{-2/3} dx = 3x^{1/3}$.

why: Dividing by a fraction multiplies by its reciprocal.

2 Evaluate at the limits: $3(8^{1/3}) - 3(1^{1/3}) = 3(2) - 3(1)$.

why: $8^{1/3} = 2$ - the limits were chosen to be perfect cubes.

3 Subtract: $6 - 3 = 3$.

why: Simple arithmetic.

Where it goes wrong

- Forgetting the factor 3 and answering 1 or 2.
- Adding one to the index the wrong way and using $x^{-5/3}$.

Takeaway $\int x^n dx = \frac{x^{n+1}}{n+1}$ works for negative and fractional n (except $n = -1$); dividing by $\frac{1}{3}$ multiplies by 3.

Question 15 FUN-09 · Functions and transformations · spec MM8.2

The graph of $y = \sin x$ is transformed to $y = 2\sin(3x) + 1$. What is the maximum value of the new function, and what is its period (in degrees)?

Answer: D

$\sin(3x)$ has period $360^\circ / 3 = 120^\circ$. $2\sin(3x)$ ranges from -2 to 2 , so $2\sin(3x) + 1$ has maximum 3 .

Full method - how and why

For $y = a\sin(bx) + c$, the amplitude, period and shift can be read straight off - as long as you know which letter does what.

- 1 The 3 inside compresses horizontally: period $= \frac{360^\circ}{3} = 120^\circ$.
why: $\sin(bx)$ completes a cycle when bx increases by 360° , i.e. when x increases by $360^\circ / b$.
- 2 The 2 outside is the amplitude: $2\sin(3x)$ ranges from -2 to 2 .
why: Multiplying the function stretches the range by that factor, symmetrically about zero.
- 3 The $+1$ raises everything: the range becomes -1 to 3 , so the maximum is 3 .
why: A vertical translation moves the whole wave, including its maximum.

Where it goes wrong

- Multiplying the period by 3 to get 1080° instead of dividing.
- Giving the amplitude 2 as the maximum and ignoring the $+1$.

Takeaway In $a\sin(bx) + c$: amplitude $|a|$, period $360^\circ / b$, midline $y = c$, maximum $c + |a|$.

Question 16 MEN-02 · Pythagoras, mensuration and 3D · spec M5.16

A cylinder has radius r and height h . Its volume is 54π and $h = 2r$. Find its total surface area.

Answer: D

$$\pi r^2 \cdot 2r = 2\pi r^3 = 54\pi \Rightarrow r = 3, h = 6. \text{ Surface area } = 2\pi r^2 + 2\pi rh = 18\pi + 36\pi = 54\pi.$$

Full method - how and why

Use the volume condition to find r , then substitute into the surface-area formula. Both formulas must be recalled.

- 1 Write the volume with $h = 2r$: $V = \pi r^2 h = \pi r^2 (2r) = 2\pi r^3$.
why: Substituting the relationship immediately reduces to one unknown.
- 2 Solve $2\pi r^3 = 54\pi$: $r^3 = 27$, so $r = 3$ and $h = 6$.
why: The π cancels, which is a good sign.
- 3 Total surface area $= 2\pi r^2$ (two circular ends) $+ 2\pi rh$ (curved surface).
why: 'Total' means include both ends; 'curved surface area' would be just $2\pi rh$.
- 4 Substitute: $2\pi(9) + 2\pi(3)(6) = 18\pi + 36\pi = 54\pi$.
why: Coincidentally equal to the volume's numerical value - a deliberate touch.

Where it goes wrong

- Giving only the curved surface 36π - an offered option.
- Forgetting that there are **two** circular ends.

Takeaway Cylinder: $V = \pi r^2 h$, curved surface $2\pi rh$, total surface $2\pi r^2 + 2\pi rh$.

Question 17 TRG-02 · Trigonometric functions and equations · spec MM4.5

Solve $2\cos^2 x + 3\sin x - 3 = 0$ for $0^\circ \leq x \leq 360^\circ$. How many solutions are there?

Answer: E

$2(1 - \sin^2 x) + 3\sin x - 3 = 0 \Rightarrow 2\sin^2 x - 3\sin x + 1 = 0 \Rightarrow (2\sin x - 1)(\sin x - 1) = 0$. $\sin x = \frac{1}{2}$: $30^\circ, 150^\circ$; $\sin x = 1$: 90° . Three solutions.

Full method - how and why

A mixed sin/cos equation. Use $\cos^2 = 1 - \sin^2$ to make it a quadratic in $\sin x$, then count solutions carefully.

- 1 Replace $\cos^2 x$ with $1 - \sin^2 x$: $2(1 - \sin^2 x) + 3\sin x - 3 = 0$.
why: The identity $\sin^2 + \cos^2 = 1$ is the only way to get a single trig function - always convert the squared one.
- 2 Tidy: $-2\sin^2 x + 3\sin x - 1 = 0$, or $2\sin^2 x - 3\sin x + 1 = 0$.
why: Multiply through by -1 so the leading coefficient is positive.
- 3 Factorise: $(2\sin x - 1)(\sin x - 1) = 0$, giving $\sin x = \frac{1}{2}$ or $\sin x = 1$.
why: Treat $\sin x$ as a single unknown.
- 4 Count solutions in $0^\circ \leq x \leq 360^\circ$: $\sin x = \frac{1}{2}$ gives 30° and 150° ; $\sin x = 1$ gives only 90° .
why: A maximum value like $\sin x = 1$ is reached just once per period, not twice - this is what makes the total odd.
- 5 Total: 3.
why: Count, do not assume 'two roots means four solutions'.

Where it goes wrong

- Assuming each value of $\sin x$ gives two solutions and answering 4.
- Sign slips when substituting the identity, producing an unfactorisable quadratic.

Takeaway Convert to one trig function, factorise, then count solutions per value - endpoints ± 1 give only one each.

Question 18 LIN-12 · Straight lines · spec MM3.1

The lines $y = mx + 4$ and $y = 3x - 2$ intersect at a point with positive x - and y -coordinates. Find the complete set of values of m .

Answer: C

$mx + 4 = 3x - 2 \Rightarrow x = \frac{6}{3-m}$, which is positive iff $m < 3$. Then $y = 3x - 2 > 0$ iff $x > \frac{2}{3}$, i.e. $\frac{6}{3-m} > \frac{2}{3}$; with $3 - m > 0$ this is $18 > 6 - 2m$, i.e. $m > -6$. So $-6 < m < 3$.

Full method - how and why

Two conditions on the intersection point. Find the intersection in terms of m , then impose positivity on each coordinate.

- 1 Find the intersection: $mx + 4 = 3x - 2 \Rightarrow x(m - 3) = -6 \Rightarrow x = \frac{6}{3-m}$.
why: Keep the denominator as $3 - m$ so that positivity is easy to read.
- 2 $x > 0$ requires $3 - m > 0$, i.e. $m < 3$.
why: The numerator 6 is positive, so the sign of x is the sign of $3 - m$.
- 3 $y > 0$: since $y = 3x - 2$, this needs $x > \frac{2}{3}$, i.e. $\frac{6}{3-m} > \frac{2}{3}$.
why: Use whichever line is simpler to express y in terms of x .
- 4 With $3 - m > 0$ we may multiply safely: $18 > 2(3 - m) \Rightarrow 18 > 6 - 2m \Rightarrow m > -6$.
why: Multiplying by a positive quantity is safe - and we already know $3 - m > 0$ from the first condition.
- 5 Combine: $-6 < m < 3$.
why: Both conditions must hold at once.

Where it goes wrong

- Checking only $x > 0$ and answering $m < 3$.
- Multiplying by $3 - m$ without noting its sign - legitimate here only because the first condition established it.

Takeaway Impose one condition at a time, and use the first to justify the algebra needed for the second.

Question 19 LOG-06 · Exponentials and logarithms · spec MM5.1

The graph of $y = 2^x$ is translated 3 units in the positive x -direction. The resulting graph has equation

Answer: A

$$y = 2^{x-3} = 2^{-3} \cdot 2^x = \frac{1}{8} \cdot 2^x. \text{ (A translation of an exponential is also a stretch.)}$$

Full method - how and why

A translation of an exponential graph is also a stretch. The algebra shows why.

- 1 Translating 3 right replaces x by $x - 3$: $y = 2^{x-3}$.

why: A shift to the right by a substitutes $x - a$.

- 2 Split the index: $2^{x-3} = 2^x \times 2^{-3} = \frac{1}{8} \cdot 2^x$.

why: $a^m \cdot a^n = a^{m+n}$ - the translation has become multiplication by a constant.

- 3 So the new graph is the old one stretched vertically by factor $\frac{1}{8}$.

why: This is special to exponentials; horizontal translation and vertical stretch are the same thing.

Where it goes wrong

- Writing $2^{x-3} = 2^x - 3$ and choosing that option.
- Using $2^{+3} = 8$ as the factor, which corresponds to a shift to the **left**.

Takeaway $a^{x-k} = a^{-k}a^x$: for exponentials, translating horizontally is the same as stretching vertically.

Question 20 CIR-05 · Circles and circle theorems · spec MM3.2

The line $y = x + c$ is a tangent to the circle $(x - 1)^2 + (y - 2)^2 = 8$. Find the possible values of c .

Answer: A

The distance from $(1, 2)$ to $x - y + c = 0$ is $\frac{|1 - 2 + c|}{\sqrt{2}} = \sqrt{8} = 2\sqrt{2}$, so $|c - 1| = 4$: $c = 5$ or $c = -3$.

Full method - how and why

Tangency means the perpendicular distance from the centre to the line equals the radius. Faster than substituting and using the discriminant.

- 1 Identify centre $(1, 2)$ and radius $\sqrt{8} = 2\sqrt{2}$.

why: Read them straight from the centre-radius form.

- 2 Write the line in the form $x - y + c = 0$ and apply the distance formula: $\frac{|1 - 2 + c|}{\sqrt{1^2 + (-1)^2}} = \frac{|c - 1|}{\sqrt{2}}$.

why: Rearranging $y = x + c$ to $x - y + c = 0$ gives $a = 1$, $b = -1$.

- 3 Set it equal to the radius: $\frac{|c - 1|}{\sqrt{2}} = 2\sqrt{2}$, so $|c - 1| = 4$.

why: Multiply both sides by $\sqrt{2}$: $2\sqrt{2} \times \sqrt{2} = 4$.

- 4 Solve the modulus: $c - 1 = \pm 4$, so $c = 5$ or $c = -3$.

why: Two tangents with that gradient exist - one on each side of the circle.

Where it goes wrong

- Dropping the modulus and finding only $c = 5$.
- Using $r = 8$ instead of $\sqrt{8}$.

Takeaway Tangency $\Leftrightarrow$ distance from centre to line = radius; the modulus always gives two answers.